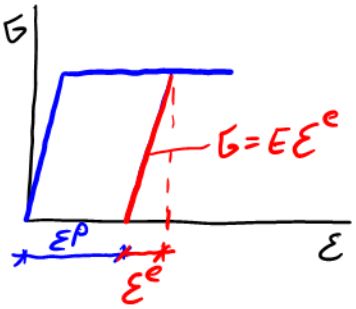
Elasto-Plasticity

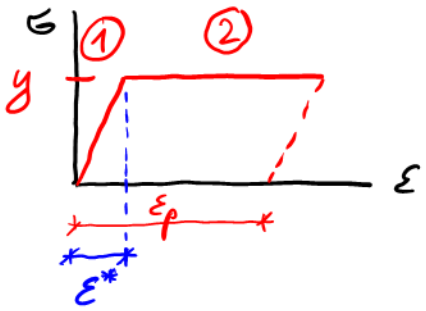
Rheological model: Prandtl-body





- Strain: $\varepsilon = \varepsilon^e + \varepsilon^p$
- Stress: $\sigma = E\varepsilon^e = E(\varepsilon - \varepsilon^p)$

Ideal Elasto-Plasticity (without Hardening)



Distinguish between domains ① / ②:

- Yield function

$$\phi(\sigma) = |\sigma| - y$$

$\phi < 0$: elast. response

$\phi = 0$: plastic yield

$\phi > 0$: not defined

$$\rightsquigarrow \boxed{\phi \leq 0}$$

- Evolution equation:

$$\dot{\varepsilon}^p = \begin{cases} 0 \text{ for } \varepsilon \leq \varepsilon^* \\ \dot{\varepsilon} \text{ for } \varepsilon > \varepsilon^* \end{cases}$$

sign of $\dot{\varepsilon}^p$ is equal to sign of σ

$$\Rightarrow \boxed{\text{Yield rule: } \dot{\varepsilon}^p = \lambda \operatorname{sign}(\sigma) = \lambda \frac{\partial \phi}{\partial \sigma}} \quad \left(\frac{\partial \phi}{\partial \sigma} = \frac{\partial |\sigma|}{\partial \sigma} = \operatorname{sign} \sigma\right) \quad (1)$$

λ is plastic parameter (consistency parameter)

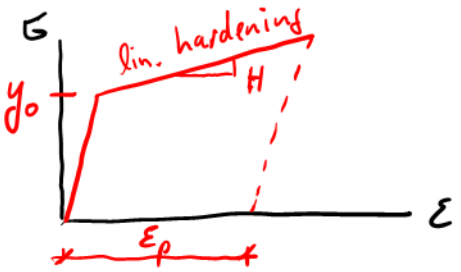
$\lambda \overset{!}{\geq} 0$, such that $\operatorname{sign} \dot{\varepsilon}^p = \operatorname{sign} \sigma$

$$\left.\begin{array}{l} \phi = 0 \rightsquigarrow \lambda > 0 \\ \phi < 0 \rightsquigarrow \lambda = 0 \end{array}\right\} \lambda\phi = 0$$

$\Rightarrow$ **Kuhn-Tucker conditions:**

$$\lambda \geq 0, \quad \phi \leq 0, \quad \lambda\phi = 0$$

Elasto-plasticity with isotropic hardening



- Yield function

$$\phi(\sigma, \alpha) = |\sigma| - \underbrace{(y_0 + \beta(\alpha))}_{y(\alpha)} \leq 0$$

$\alpha \hat{=}$ internal variable measuring intensity of plastic strain

- Special case linear hardening:

$$\phi(\sigma, \alpha) = |\sigma| - (y_0 + H\alpha) \leq 0$$

H ↳ hardening modulus

- Evolution equation for internal variable:

$$\dot{\alpha} = |\dot{\varepsilon}^p| \quad \text{with} \quad \dot{\varepsilon}^p = \lambda \operatorname{sign}(\sigma)$$

$$= \lambda \geq 0$$

- λ is computed from consistency condition $\dot{\phi} = 0$ (for plastic yielding)

$$\dot{\phi}(\sigma, \alpha) = \frac{\partial \phi}{\partial \sigma}\dot{\sigma} + \frac{\partial \phi}{\partial \alpha}\dot{\alpha} = \underbrace{\frac{\partial |\sigma|}{\partial \sigma}}_{\text{sign}(\sigma)}\dot{\sigma} + \frac{\partial \phi}{\partial \alpha}\dot{\alpha}$$

$\dot{\sigma} = E(\dot{\varepsilon} - \dot{\varepsilon}^p)$

$$= \text{sign}(\sigma) E\dot{\varepsilon} - \text{sign}(\sigma) E\lambda \,\text{sign}(\sigma) + \underbrace{\frac{\partial \phi}{\partial \alpha}}\lambda$$

$$\stackrel{!}{=} 0$$

depends in general on α
$\rightsquigarrow$ unknowns λ and α
$\rightsquigarrow$ compute λ and α from $\dot{\phi} = 0$ and time integration of $\dot{\alpha} = \lambda$

However, for linear hardening we have $\frac{\partial \phi}{\partial \alpha} = -H$ (independent from α !)

$$\rightsquigarrow \dot{\phi}(\sigma, \lambda) = \text{sign}(\sigma) E\dot{\varepsilon} - \lambda(E + H) \stackrel{!}{=} 0$$

$$\Leftrightarrow \quad \lambda = \frac{E}{E+H}\,\text{sign}(\sigma)\,\dot{\varepsilon}$$

$$\Rightarrow \quad \dot{\varepsilon}^p = \lambda\,\text{sign}(\sigma) = \frac{E}{E+H}\,\dot{\varepsilon}$$

$$\dot{\varepsilon}^e = \dot{\varepsilon} - \dot{\varepsilon}^p = \left(\lambda - \frac{E}{E+H}\right)\dot{\varepsilon} = \frac{H}{E+H}\,\dot{\varepsilon}$$

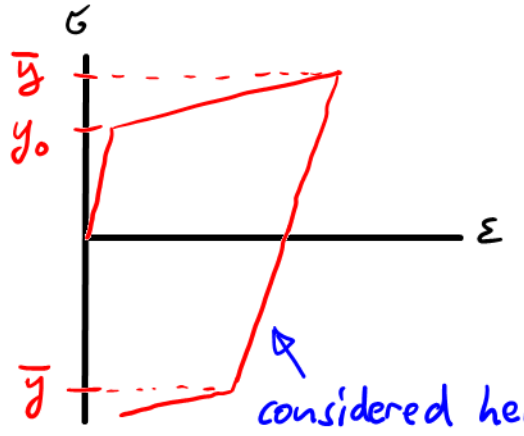
(independent from $\alpha \rightsquigarrow \alpha$ not needed)

insert in $\dot{\sigma} = E\dot{\varepsilon}^e = \underbrace{\frac{EH}{E+H}}\,\dot{\varepsilon}$

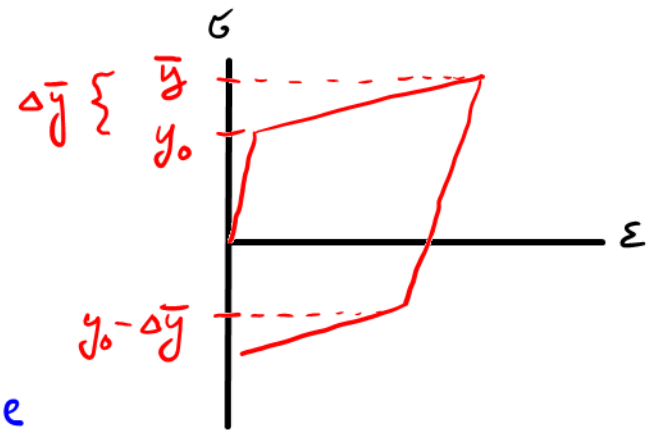
$=: E^{ep}$ in case of plastic yield, otherwise $E^{ep} = E^{el}$

Elasto-Plasticity with isotropic von Mises Hardening in 3D

Isotropic hardening



Kinematic hardening



- Additive split of strains:

$$\underline{\varepsilon} = \underline{\varepsilon}^e + \underline{\varepsilon}^p \quad \Leftrightarrow \quad \underline{\varepsilon}^e = \underline{\varepsilon} - \underline{\varepsilon}^p$$

- Internal variable for isotropic hardening: α → scalar

- Energy:

$$\psi = \psi(\underline{\varepsilon}, \underline{\varepsilon}^p, \alpha) = \psi^e(\underline{\varepsilon}^e = \underline{\varepsilon} - \underline{\varepsilon}^p) + \psi^p(\alpha)$$

$$\rightsquigarrow \quad \dot{\psi} = \frac{\partial \psi}{\partial \underline{\varepsilon}} \cdot \dot{\underline{\varepsilon}} + \frac{\partial \psi}{\partial \underline{\varepsilon}^p} \cdot \dot{\underline{\varepsilon}}^p + \frac{\partial \psi}{\partial \alpha} \dot{\alpha}$$

$$\frac{\partial \psi}{\partial \underline{\varepsilon}^e} : \frac{\partial \underline{\varepsilon}^e}{\partial \underline{\varepsilon}} = \frac{\partial \psi}{\partial \underline{\varepsilon}^e} \qquad \frac{\partial \psi}{\partial \underline{\varepsilon}^e} : \frac{\partial \underline{\varepsilon}^e}{\partial \underline{\varepsilon}^p} = -\frac{\partial \psi}{\partial \underline{\varepsilon}^e}$$

insert in Clausius-Duhem inequality

$$\underline{\sigma} \cdot \dot{\underline{\varepsilon}} - \dot{\psi} = \underline{\sigma} \cdot \dot{\underline{\varepsilon}} - \frac{\partial \psi}{\partial \underline{\varepsilon}^e} \cdot \dot{\underline{\varepsilon}} + \frac{\partial \psi}{\partial \underline{\varepsilon}^e} \cdot \dot{\underline{\varepsilon}}^p - \frac{\partial \psi}{\partial \alpha} \dot{\alpha}$$

$$= \left(\underline{\sigma} - \frac{\partial \psi}{\partial \underline{\varepsilon}^e}\right) \cdot \dot{\underline{\varepsilon}} + \frac{\partial \psi}{\partial \underline{\varepsilon}^e} \cdot \dot{\underline{\varepsilon}}^p - \frac{\partial \psi}{\partial \alpha} \dot{\alpha} \geq 0$$

Standard argument of rational continuum mechanics:

$$\rightsquigarrow \quad \underline{\sigma} = \frac{\partial \psi}{\partial \underline{\varepsilon}^e} = \frac{\partial \psi^e(\underline{\varepsilon}^e)}{\partial \underline{\varepsilon}^e}$$

$$\rightsquigarrow \quad \mathcal{D}_{int} := \underline{\sigma} \cdot \dot{\underline{\varepsilon}}^p - \beta \dot{\alpha} \geq 0$$

β: thermodynamic force

$$\beta := \frac{\partial \psi}{\partial \alpha} = \frac{\partial \psi^p(\alpha)}{\partial \alpha}$$

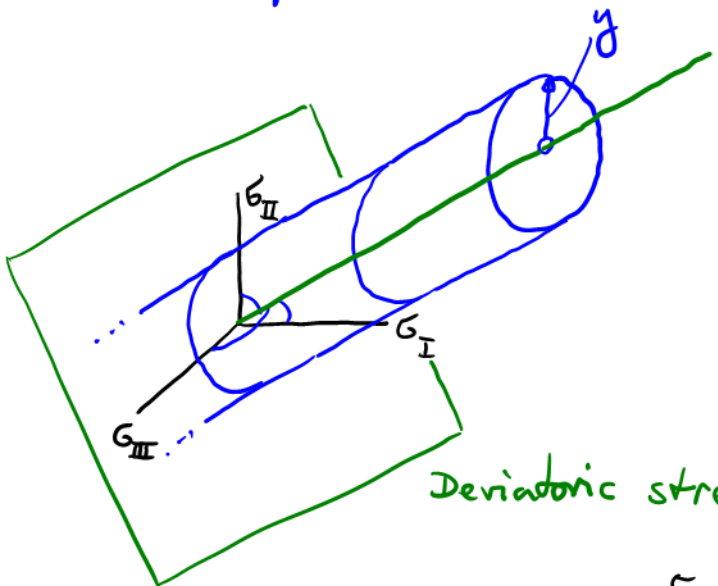
- Elastic domain in stress space:

$$E_\sigma := \left\{ \underline{\sigma} \in \mathbb{R}^6 \mid \phi \leq 0 \right\}$$

ϕ: yield "surface"

Von Mises yield surface

In metals, plastic strains are nearly incompressible
$\rightsquigarrow$ "metal plasticity"



hydrostatic axis σ_{hyd}

$$\underline{\sigma} = \sigma_{hyd} + \operatorname{dev} \underline{\sigma}$$

with $\sigma_{hyd} = \underbrace{\frac{1}{3}(\operatorname{tr} \underline{\sigma})}_{-p \text{ (pressure)}} \underline{1}$

Deviatoric stress plane $\operatorname{dev} \underline{\sigma}$

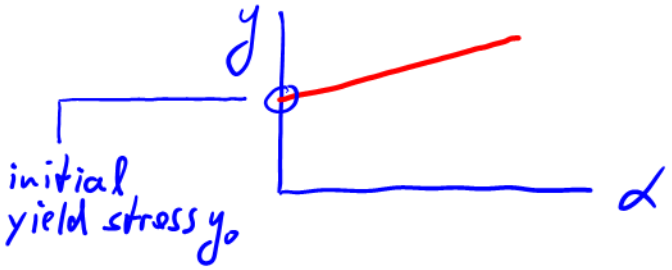
⇒ in case of von Mises hardening, "J_2-plasticity" is considered, ie. second invariant of stress is considered as comparative stress measure:

$$J_2[\underline{\sigma}] = \|\operatorname{dev}\underline{\sigma}\|$$

$$\Rightarrow \quad \phi := J_2 - y(\alpha)$$

$$= \|\operatorname{dev}\underline{\sigma}\| - y(\alpha)$$

characterizes yield surface, which enlarges as a result of plastic strains



Identification of y_0 from uniaxial tension test:

let tension direction of experiment be x_1-direction, then

$$\operatorname{dev}\underline{\sigma} = \underline{\sigma} - \tfrac{1}{3}(\operatorname{tr}\underline{\sigma})\underline{1} = \begin{bmatrix}\sigma_{11} & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix} - \tfrac{1}{3}\sigma_{11}\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$= \begin{bmatrix}\frac{2}{3}\sigma_{11} & 0 & 0\\ 0 & -\frac{1}{3}\sigma_{11} & 0\\ 0 & 0 & -\frac{1}{3}\sigma_{11}\end{bmatrix}$$

$$\| \operatorname{dev} \underline{\sigma} \| = \sqrt{\left(\tfrac{2}{3}\sigma_{11}\right)^2 + 2\left(-\tfrac{1}{3}\sigma_{11}\right)^2} = \sqrt{\left(\tfrac{4}{9} + \tfrac{2}{9}\right)\sigma_{11}^2}$$

$$= \sqrt{\tfrac{2}{3}}\,\sigma_{11}$$

σ_{11}: value in experiment, where hardening begins



$\Rightarrow$ then we have $\| \operatorname{dev} \sigma \| = \sqrt{\tfrac{2}{3}}\, y_0$

$\Longrightarrow$ Yield criterion:

$$\phi := \| \operatorname{dev} \underline{\sigma} \| - \sqrt{\tfrac{2}{3}}\,(y_0 + \beta) \leq 0$$

$\psi^p(\alpha)$ must be constructed such that $\beta = \frac{\partial \psi}{\partial \alpha}$ describes experimental hardening curve, e.g. linear hardening as simplest approximation:

$$\psi^p(\alpha) = \tfrac{1}{2} H \alpha^2 \rightsquigarrow \beta = \frac{\partial \psi}{\partial \alpha} = H\alpha$$

Evolution equations

- Associative plasticity:

  $\underline{\dot{\varepsilon}}^p \rightsquigarrow$ yield direction is orthogonal to yield surface

- Principle of maximal dissipation
  "Out of all admissible stress states, the true stresses maximize dissipation" (Hill 1948)
  ↳ Optimization problem with constraint condition $\phi \leq 0$

$$\rightsquigarrow \mathcal{L} := -\mathcal{D}_{int}(\underline{\sigma}, \beta) + \underbrace{\lambda \phi(\underline{\sigma}, \beta)} \rightarrow \text{stat.}$$

incorporation of constraint by Lagrange multiplier $\lambda \geq 0$

↳ Kuhn-Tucker method
↳ ϕ must be convex
(ok for von Mises due to cylinder in stress space)

$$\Rightarrow \mathcal{L}(\underline{\sigma}, \beta, \lambda) = -\underline{\sigma} \cdot \underline{\dot{\varepsilon}}^p + \beta \dot{\alpha} + \lambda \phi(\underline{\sigma}, \beta) \rightarrow \text{stat.}$$

$$\rightsquigarrow \frac{\partial \mathscr{L}}{\partial \underline{\sigma}} = -\dot{\underline{\varepsilon}}^p + \lambda \frac{\partial \phi}{\partial \underline{\sigma}} \stackrel{!}{=} 0$$

$$\Leftrightarrow \boxed{\dot{\underline{\varepsilon}}^p = \lambda \frac{\partial \phi}{\partial \underline{\sigma}} = \lambda \underline{n}}$$ "Yield/flow rule"

$$\rightsquigarrow \frac{\partial \mathscr{L}}{\partial \beta} = \dot{\alpha} + \lambda \frac{\partial \phi}{\partial \beta} \stackrel{!}{=} 0$$

$$\Leftrightarrow \boxed{\dot{\alpha} = -\lambda \frac{\partial \phi}{\partial \beta} = \lambda \sqrt{\frac{2}{3}}}$$ Evolution of internal variable

$$\rightsquigarrow \frac{\partial \mathscr{L}}{\partial \lambda} = \phi \stackrel{!}{=} 0 \quad \checkmark \text{ (in case of plastic flow)}$$